Discussion of “Ranking Algorithms and Equilibrium Prices” by Huang, Korganbekova, and Korganbekova

Marketing-Industrial Organization Conference
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Comments:

1. Demand elasticity tests
2. Exploring heterogeneity

Overview

Better reputation
$\bar{s} \uparrow$ (exogenous)
$\Rightarrow$ better rank

Key observation from experiment

Lower price

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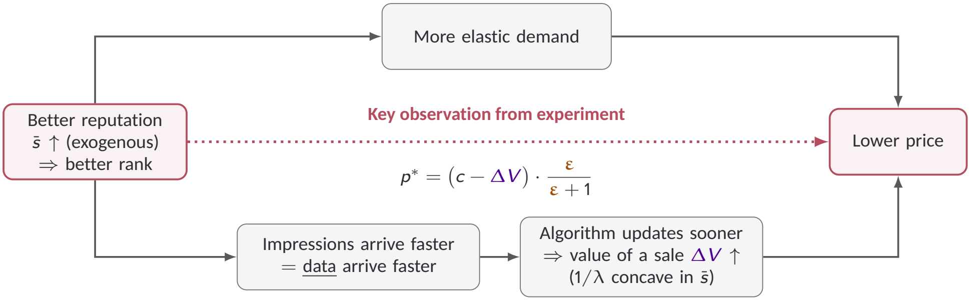
Key observation from experiment

$$p^* = (c - \Delta V) \cdot \frac{\varepsilon}{\varepsilon + 1}$$

Lower price

Overview

1. Static demand channel



2. Dynamic incentive: invest in future reputation

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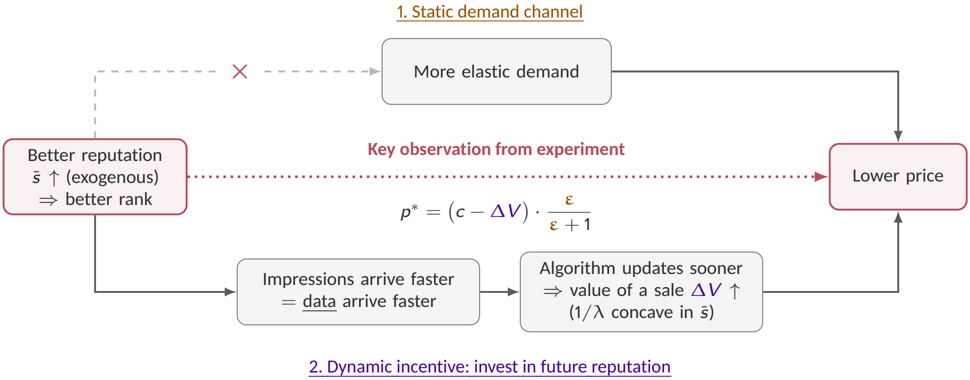
1. Static demand channel
More elastic demand
Key observation from experiment
Better reputation
$\bar{s} \uparrow$ (exogenous)
$\Rightarrow$ better rank
Lower price
$p^* = (c - \Delta V) \cdot \frac{\varepsilon}{\varepsilon + 1}$
Impressions arrive faster
= data arrive faster
Algorithm updates sooner
$\Rightarrow$ value of a sale $\Delta V \uparrow$
($1/\lambda$ concave in $\bar{s}$)
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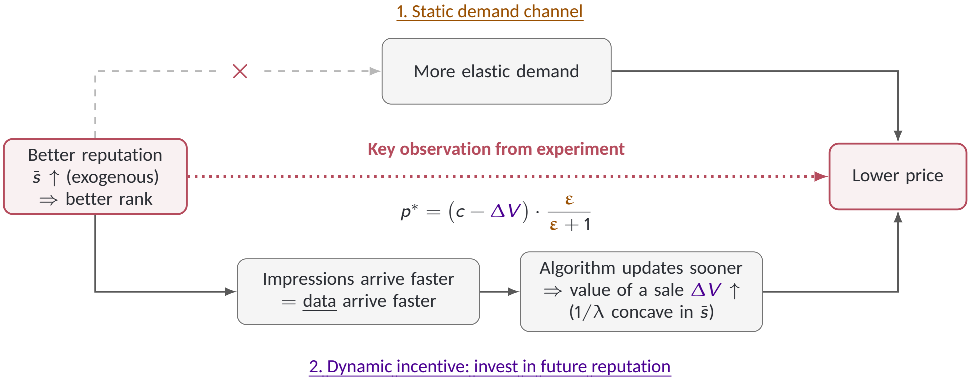

Reputation is easier to maintain than to rebuild

- Traffic is the reward and the data the algorithm learns from
- Better reputation raises the **return** to investing and the **capacity** to invest
- Asymmetry in learning: self-correcting above, self-sealing below

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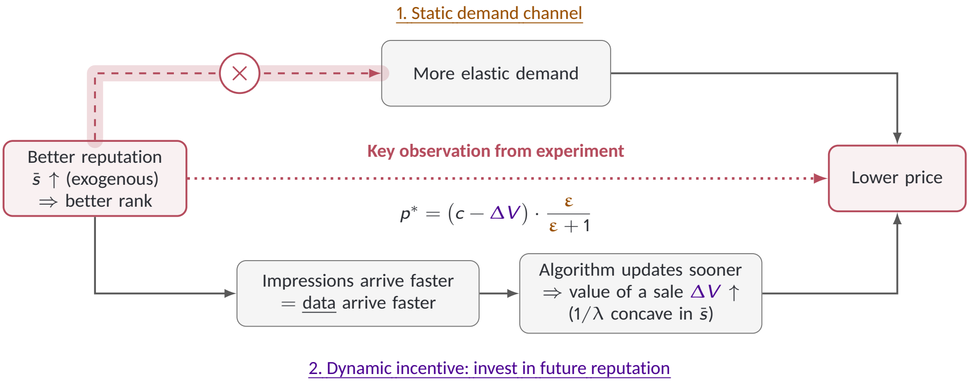

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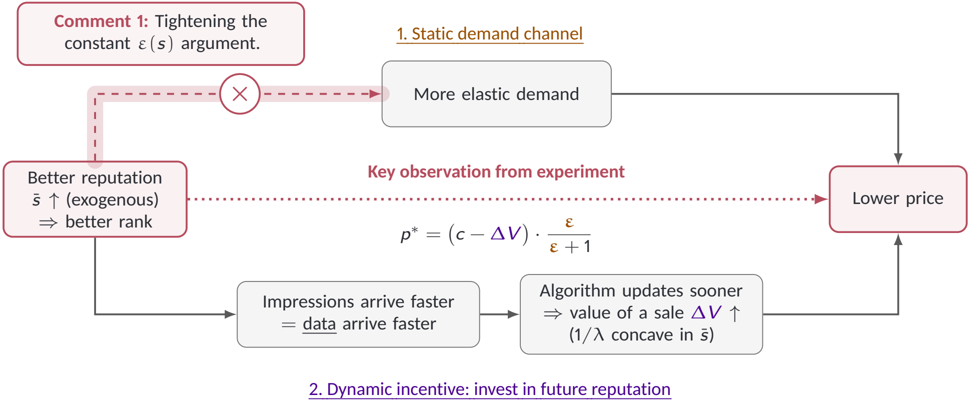

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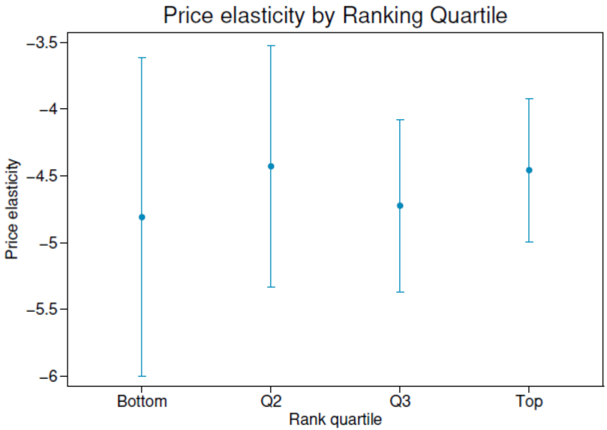

Figure 6: Estimated Price Elasticity by Rank Quartile

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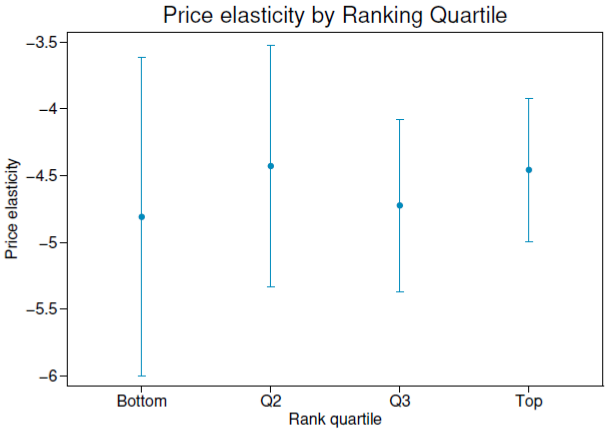

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- If experimental rank improvement makes $|\epsilon| = 4.5 \rightarrow 4.9$, that's enough to rationalize 2.3% decrease in prices.

Comment 1: Demand elasticity

Two concrete suggestions

1. A more parsimonious functional form, e.g.,

$$\mathbb{E}\left[q_{jt} \mid p_{jt}, \mu_j, \delta_{gt}\right] = \exp\left(\alpha_1 \log\left(p_{jt}\right) + \alpha_2 \log\left(p_{jt}\right) \times \log\left(\text{rank}_{jt}\right) + \mu_j + \delta_{gt}\right)$$

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2. Instead, run tests of the form: $H_0 : \alpha_2 \geqslant A$.

   — Then ask, what is the largest value A such that you fail to reject? Call this $A_{\text{worst case}}$.

   — If $\alpha_2 = A_{\text{worst case}}$, what proportion of the 2.3% price decrease would the demand elasticity channel explain?

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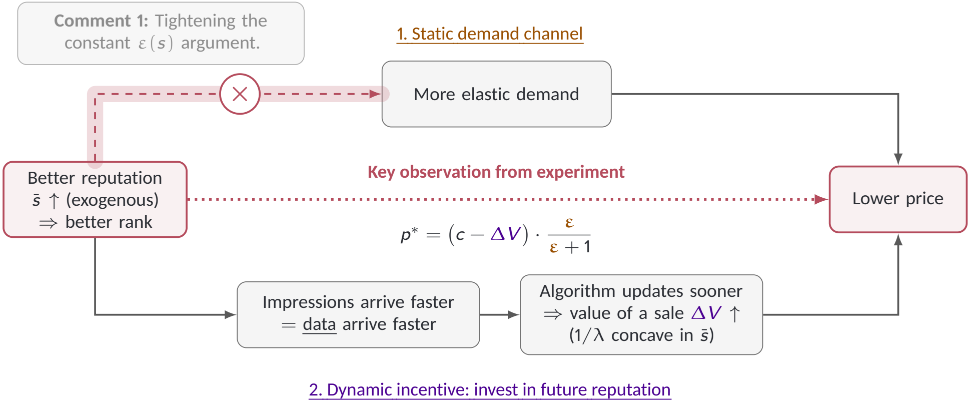

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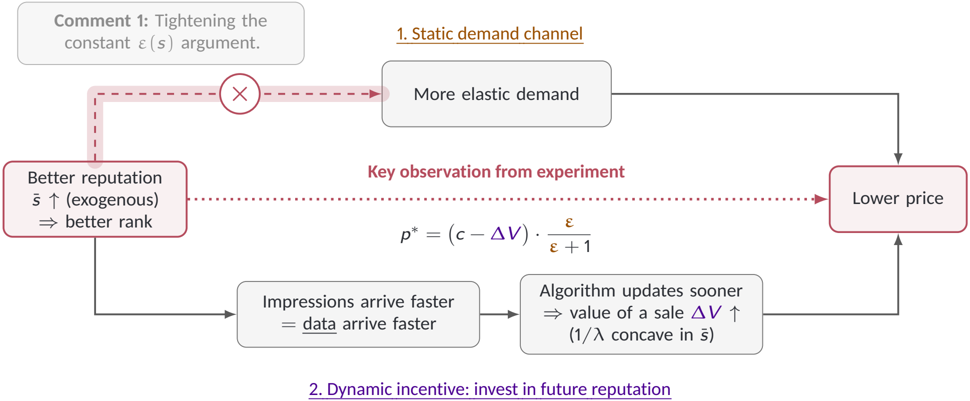

Reputation is easier to maintain than to rebuild

Comment 2: Does everyone want to maintain? Unpacking heterogeneity.

— Traffic is the reward and the data the algorithm learns from

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Comment 2: Unpacking heterogeneity

“Reputation easier to maintain than to rebuild.” But does everyone want to maintain?

- Positive price responses for some product clusters: Which clusters are these?

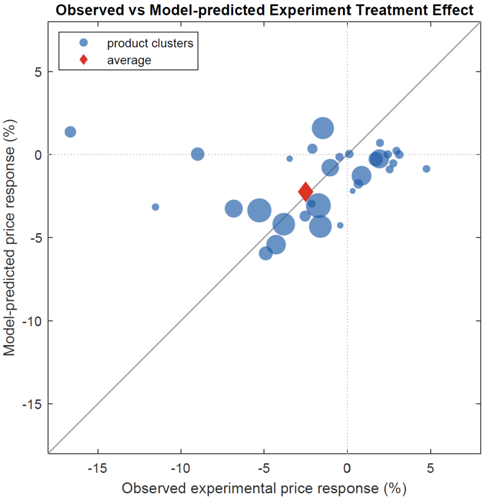

Figure B3: Model validation by cluster: predicted and observed price responses

Other questions & comments

- Optimistic prior would seem to incentivize creation of “new” products (maybe just existing products with some slight tweak). Evidence of that? Platform rules prohibiting it?

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Closing thoughts

- Clean experiment, and the mechanism is the right one.
- Two asks:
  - Demand elasiticty: How much of the price response can a gradient in ε deliver?
  - Heterogeneity: For *which* products (clusters) does the story hold?
- Can you say more empirically about the effects of the **self-correcting above, self-sealing below** asymmetry?

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Structural model:

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    - $\rightarrow$ $F(\hat{s}_{jt-1})$ is a *descendant of past prices* — the investment mechanism itself.
  - Grouped FEs (Bonhomme, Lamadon, and Manresa (2022)) likely absorb ξ, so I'm not actually too worried about this in the structural model.